

Week 7: Kinship

SOC6708 ADA

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In this lab we will be calculating surviving daughters and mothers using the GKP equations, and also exploring matrix-based models using DemoKin.

Load in data

Note you will have to install the DemoKin package.

```
library(tidyverse)
library(readxl)
library(DemoKin)
```

For the GKP we will be using fertility and mortality rates from WPP; we also need to load in the file that has summary indicators. You should have these files from previous weeks.

```
df <- read_csv("../data/WPP2024_Fertility_by_Age5.csv")
dm <- read_csv("../data/WPP2024_Life_Table_Abridged_Medium_1950-2023.csv")
di <- read_csv("../data/WPP2024_Demographic_Indicators_Medium.csv")
```

For the matrix models, we will use fertility and mortality rates by single year of age. I've already cleaned these up (the file is in the repo).

```
ds <- read_csv("../data/wpp_pop_fx_px_female.csv")
```

Goodman-Keyfitz-Pullum equations

Surviving daughters

Let's look at the surviving daughters in Kenya over time.

First, clean up the fertility and mortality data:

```
fert <- df |>
  filter(Location == "Kenya") |>
  select(Time, AgeGrp, ASFR) |>
  mutate(age = as.numeric(str_remove(AgeGrp, "-.*")))

mort <- dm |>
  filter(Location == "Kenya", Sex == "Female") |>
  select(Time, AgeGrp, lx) |>
  filter(AgeGrp!="1-4")|> # don't need this
  mutate(age = as.numeric(str_remove(AgeGrp, "-.*")))
```

Exercise

Do a couple of graphs to show how fertility and mortality have changed over time.

Let's calculate expected number of surviving daughters over time for a 60 year old:

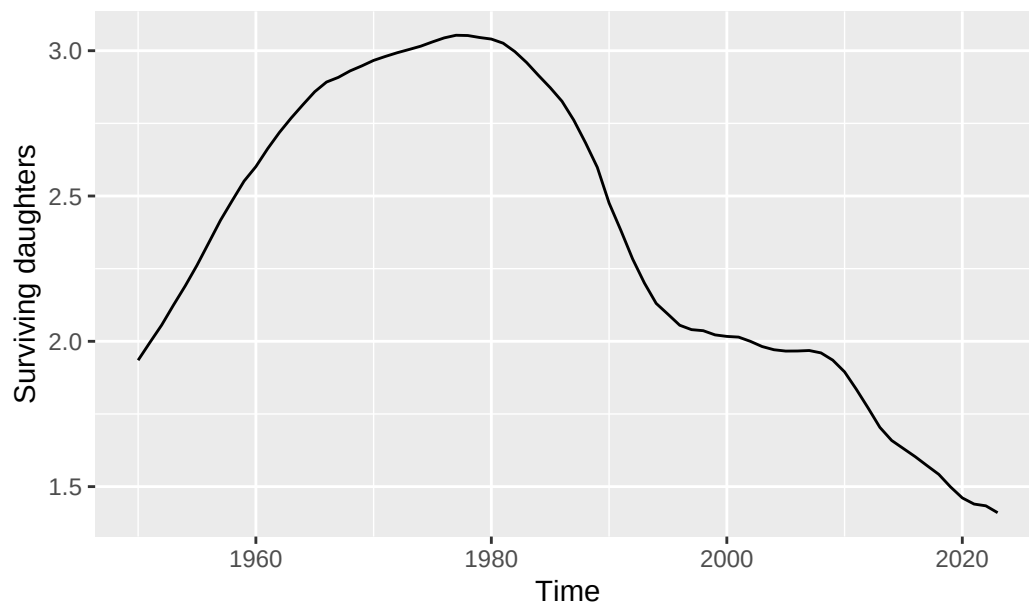
```
a <- 60

surviving_daughters <- fert |>
  mutate(shifted_age = a - age) |>
  left_join(mort |> rename(shifted_age = age) |>
            select(-AgeGrp) ) |>
  mutate(prod = ASFR*5*0.4886/1000*lx/100000) |>
  group_by(Time) |>
  summarize(daughters = sum(prod)) |>
  filter(Time < 2024)
```

Plot this:

```
surviving_daughters |>
  ggplot(aes(Time, daughters)) +
  geom_line()+
  labs(y = "Surviving daughters", title = "Expected number of surviving daughters for a 60 y
```

Expected number of surviving daughters for a 60 year woman,



Exercise

Compare the trajectory above to the trajectory of TFR for Kenya. Explain the similarities / differences.

Surviving mothers

To calculate the probability that a mother is still alive, we need to calculate W_x , which is the age distribution of mothers. We'll calculate this using stable population assumptions as in the lecture notes, which requires first calculating the stable growth rate r . I'm using the approximation from week 5, i.e.

$$r \approx \frac{NRR}{\mu}$$

where μ is the mean age at childbearing.

```
dr <- di |>
  filter(Location=="Kenya") |>
  select(Time, NRR, MAC) |>
  mutate(r = NRR/MAC)

Wx <- mort |>
  left_join(fert) |>
```

```

mutate(ASFR = ifelse(is.na(ASFR), 0, ASFR)) |>
mutate(age = ifelse(is.na(age), 100, age)) |>
mutate(prod = ASFR*5*0.4886/1000*1x/100000) |>
left_join(dr) |>
mutate(dist = exp(-1*r*age)*prod) |>
group_by(Time) |>
mutate(sum = sum(dist)) |>
mutate(prop = dist/sum) # renormalize

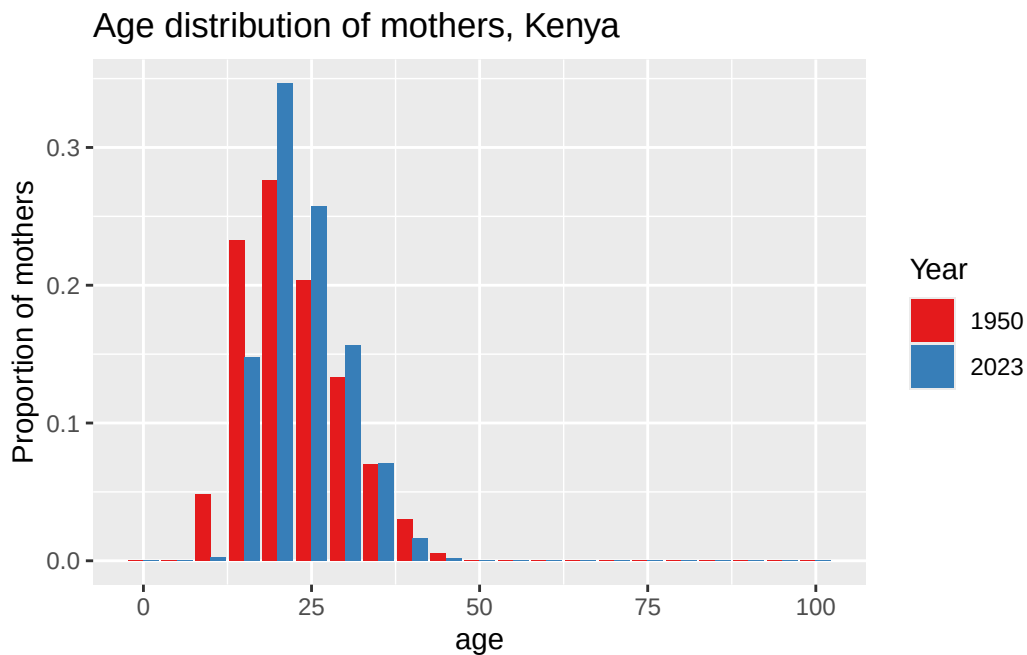
```

Plot a couple of years to make sure it broadly makes sense

```

Wx |>
filter(Time == 2023|Time == 1950) |>
ggplot(aes(age, prop, fill = factor(Time))) +
geom_bar(stat = "identity", position = 'dodge')+
scale_fill_brewer(palette = "Set1", name = "Year")+
labs(y = "Proportion of mothers", title = "Age distribution of mothers, Kenya")

```



Now we have this age distribution, we can calculate the probability that a mother is alive. For this one, let's use ego age = 45.

```

a <- 45

age_shifted = seq(0, 100, by = 5) + a

# shift mortality rates
mort_shift <- mort |>
  mutate(age = ifelse(is.na(age), 100, age)) |>
  filter(age %in% age_shifted) |>
  rename(lx_shifted = lx, age_shifted = age) |>
  select(-AgeGrp)

surviving_mothers <- mort |>
  mutate(age_shifted = age + a) |>
  left_join(mort_shift) |>
  mutate(lx_shifted = ifelse(is.na(lx_shifted), 0, lx_shifted)) |>
  left_join(fert) |>
  mutate(ASFR = ifelse(is.na(ASFR), 0, ASFR)) |>
  left_join(Wx |>
    select(Time, AgeGrp, age, prop)) |>
  mutate(ratio = lx_shifted/lx) |>
  mutate(prod = prop*ratio) |>
  drop_na(age) |>
  group_by(Time) |>
  summarize(M1 = sum(prod))

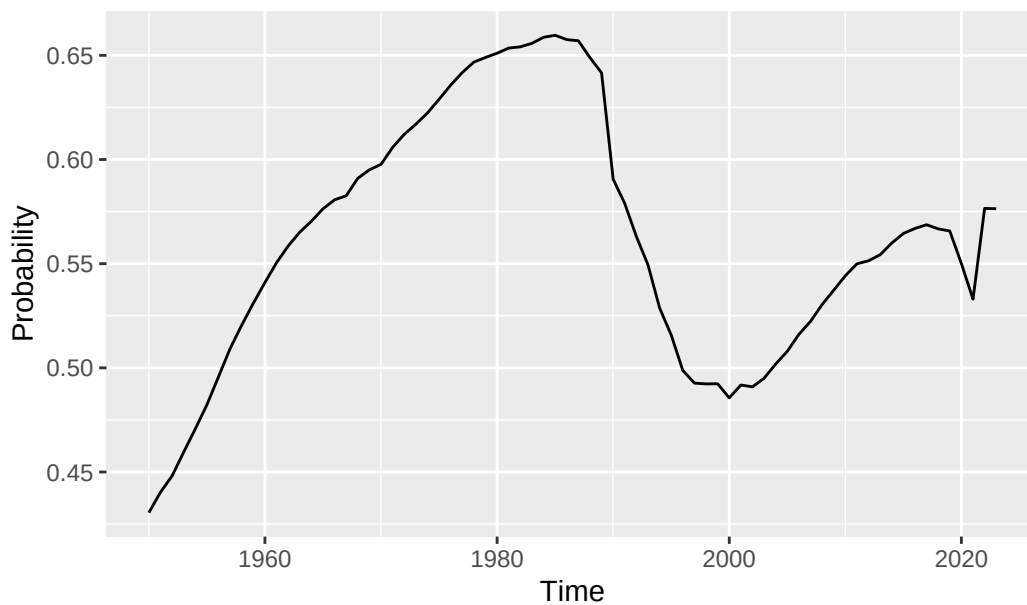
```

```

surviving_mothers|>
  ggplot(aes(Time, M1)) +
  geom_line()+
  labs(y = "Probability", title = "Probability that mother is still alive for Kenyan woman a

```

Probability that mother is still alive for Kenyan woman aged 45



Question for you

How would you convert this probability into the number of 45 women in Kenya who have lost their mother (say, in 2023)?

DemoKin

Now we will use the **DemoKin** package to calculate surviving kin of different types (female only) in Canada. This code replicates the plots in the lecture notes.

Let's get the fertility and mortality rates in the right format:

```
asfr_2023 <- ds |>
  filter(region=="Canada") |>
  filter(year==2023) |>
  select(age, year, fx) |>
  mutate(fx = fx/1000) |>
  pivot_wider(names_from = year, values_from = fx) |>
  select(-age) |>
  as.matrix()

surv_2023 <- ds |>
```

```

filter(region=="Canada") |>
filter(year==2023) |>
select(age, year, px) |>
pivot_wider(names_from = year, values_from = px) |>
select(-age) |>
as.matrix()

```

The main function in the DemoKin package is then `kin`, which takes an inputs fertility and survival matrices.

```
kin_2023 <- kin(p = surv_2023, f = asfr_2023, time_invariant = TRUE)
```

This object has two elements, a full description and a summary.

```
kin_2023$kin_summary
```

```

# A tibble: 1,414 x 9
  age_focal kin    year count_living mean_age sd_age count_dead count_cum_dead
  <int> <chr> <lgl>      <dbl>    <dbl> <dbl>    <dbl>      <dbl>
1      0 coa  NA      0.153     8.59  6.33  0.0000513  0.0000513
2      0 cya  NA      0.0491    4.21  3.84  0.0000333  0.0000333
3      0 d    NA      0         NaN   NaN    0          0
4      0 gd   NA      0         NaN   NaN    0          0
5      0 ggd  NA      0         NaN   NaN    0          0
6      0 ggm  NA      0.301    86.2   6.66  0.0274     0.0274
7      0 gm   NA      0.934    62.2   7.26  0.00613    0.00613
8      0 m    NA      1        31.3   5.20  0.000500   0.000500
9      0 nos  NA      0.0000656 1.45   1.75  0.000000102 0.000000102
10     0 nys  NA      0         NaN   NaN    0          0
# i 1,404 more rows
# i 1 more variable: mean_age_lost <dbl>

```

Note that this `kin` column refers to different types of kin, to make this more readable we can use the `rename_kin` function:

```

kin_2023$kin_summary |>
  rename_kin()

```

```

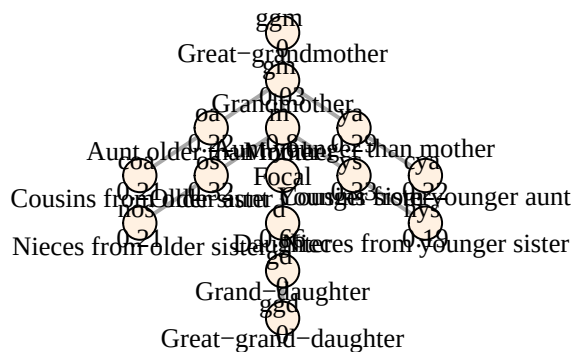
# A tibble: 1,414 x 10
  age_focal kin    year count_living mean_age sd_age count_dead count_cum_dead

```

[illegible]

```
kin_2023$kin_summary |>
  rename_kin() |>
  filter(kin_label == "Daughters", age_focal==60)
```

We can plot a tree summary for a 45 year old:

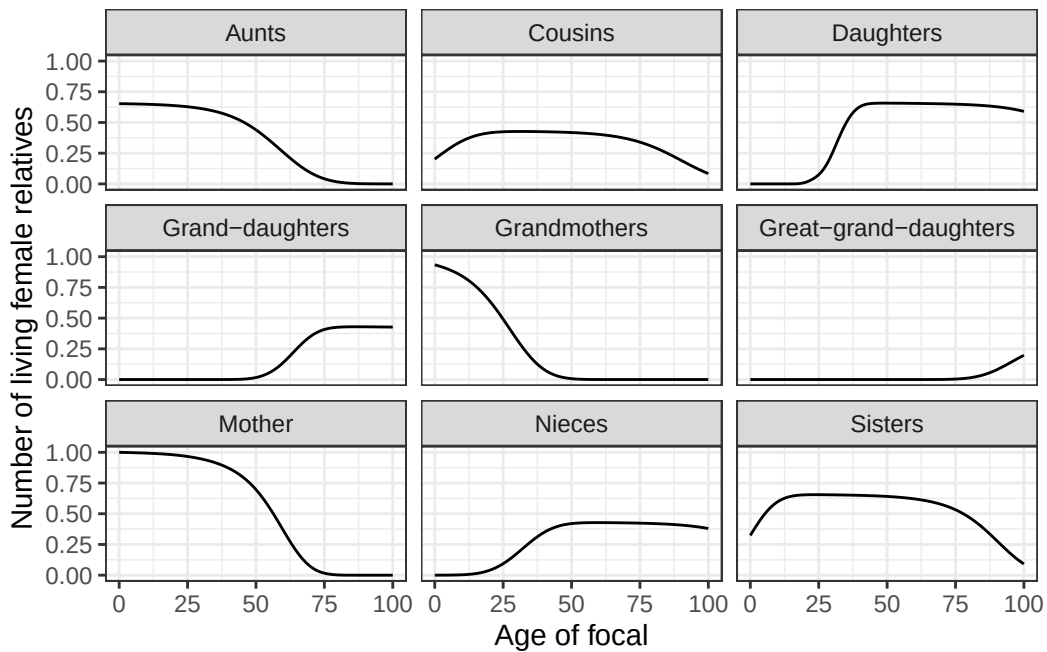


We can specify the types of kin we are interested in.

```
kin_2023 <-  
  kin(  
    p = surv_2023  
    , f = asfr_2023  
    , output_kin = c("c", "d", "gd", "ggd", "gm", "m", "n", "a", "s")  
    , time_invariant = TRUE  
  )
```

Then plot the number surviving over age

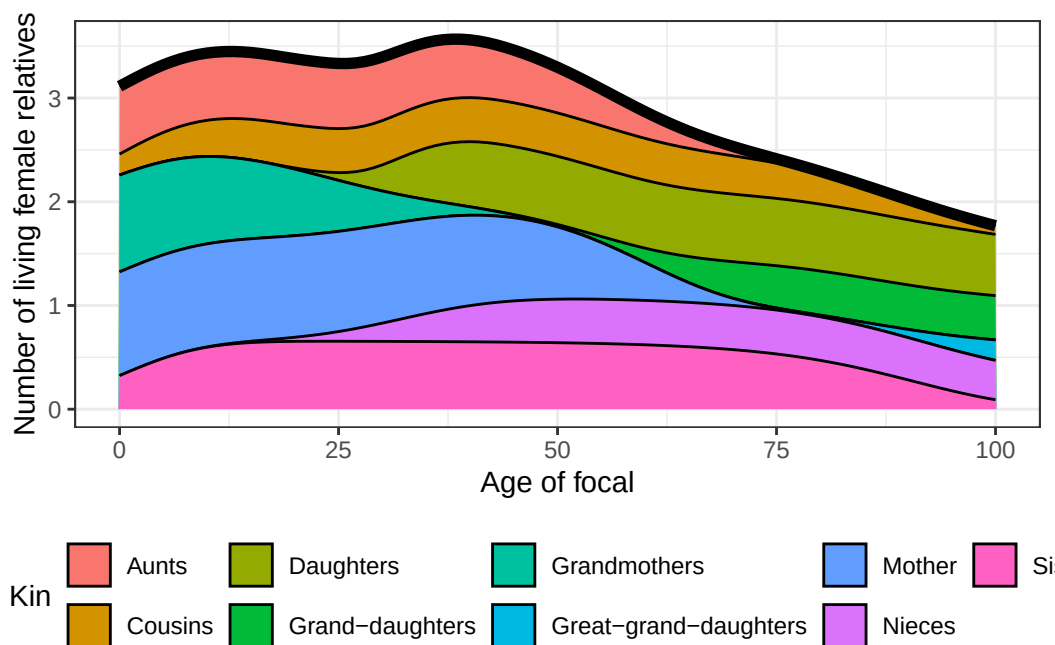
```
kin_2023$kin_summary |>  
  rename_kin() |>  
  ggplot() +  
  geom_line(aes(age_focal, count_living)) +  
  theme_bw() +  
  labs(x = "Age of focal", y = "Number of living female relatives") +  
  facet_wrap(~kin_label)
```



Combine these counts:

```
counts <-
  kin_2023$kin_summary |>
  group_by(age_focal) |>
  summarise(count_living = sum(count_living)) |>
  ungroup()

kin_2023$kin_summary |>
  select(age_focal, kin, count_living) |>
  rename_kin() |>
  ggplot(aes(x = age_focal, y = count_living)) +
  geom_area(aes(fill = kin_label), colour = "black") +
  geom_line(data = counts, linewidth = 2) +
  labs(x = "Age of focal",
       y = "Number of living female relatives",
       fill = "Kin") +
  theme_bw() +
  theme(legend.position = "bottom")
```



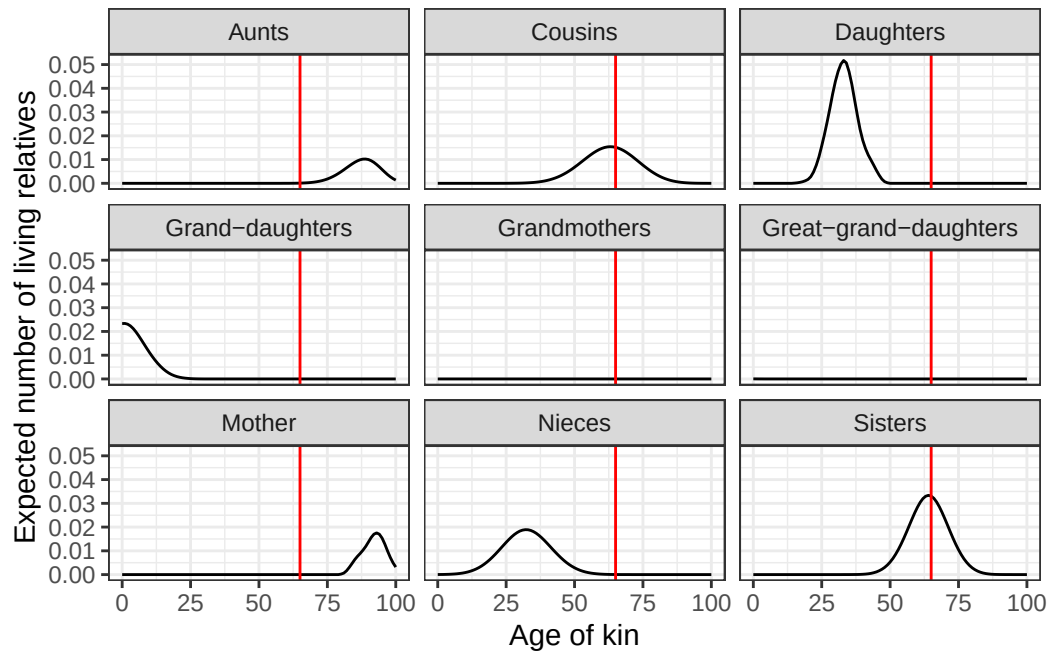
Finally, we can plot the age distribution of relatives for Canadian women aged 65.

```
kin_2023$kin_full |>
  rename_kin() |>
  filter(age_focal == 65) |>
  ggplot(aes(age_kin, living)) +
```

```

geom_line() +
geom_vline(xintercept = 65, colour = "red") +
labs(x = "Age of kin",
     y = "Expected number of living relatives") +
theme_bw() +
facet_wrap(~kin_label)

```



Can repeat the same exercise using time varying rates:

```

asfr <- ds |>
  filter(region=="Canada") |>
  select(age, year, fx) |>
  mutate(fx = fx/1000) |>
  pivot_wider(names_from = year, values_from = fx) |>
  select(-age) |>
  as.matrix()

surv <- ds |>
  filter(region=="Canada") |>
  select(age, year, px) |>
  pivot_wider(names_from = year, values_from = px) |>
  select(-age) |>
  as.matrix()

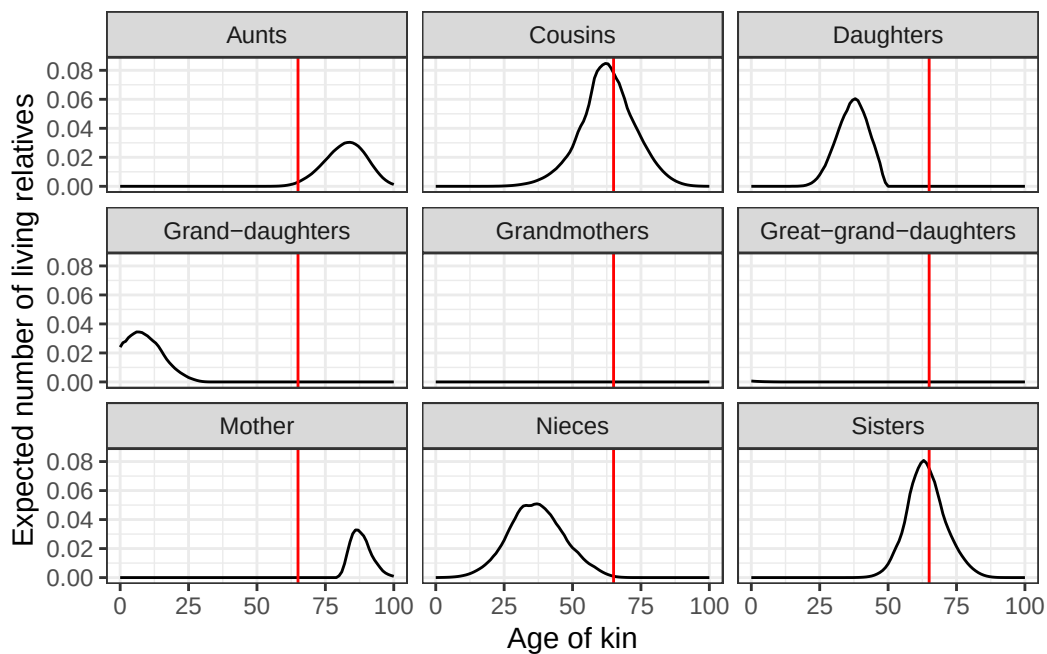
```

```

kin_tv <- kin(p = surv, f = asfr, time_invariant = FALSE,
             output_kin = c("c", "d", "gd", "ggd", "gm", "m", "n", "a", "s"))

kin_tv$kin_full |>
  filter(year==2023) |>
  rename_kin() |>
  filter(age_focal == 65) |>
  ggplot(aes(age_kin, living)) +
  geom_line() +
  geom_vline(xintercept = 65, colour = "red") +
  labs(x = "Age of kin",
       y = "Expected number of living relatives") +
  theme_bw() +
  facet_wrap(~kin_label)

```



Exercise

Pick another country to get the kinship structure for women. Compare to Canada and briefly discuss similarities and differences.